EXTENSION WORKBOOK

SWIM & WATER SAFETY

Health & Physical Education | Year 7-8 | Extension & Enrichment

This workbook goes deeper on every topic.
Analyse, evaluate, justify, and use evidence.
Research is required for some questions.

STUDENT NAME

CLASS

TEACHER

Extension Workbook — Expectations:

Answers should go beyond description. Analyse, evaluate, justify, and use evidence. Where a question asks "why," give a reasoned explanation with supporting data. Research questions require a source URL.

1 Statistical Analysis — Victorian & Global Drowning Data

Advanced Context

- 54 fatal drownings in Victoria 2023-24 — highest in over a decade
- Victoria's population approx. 6.9 million — drowning rate = 0.78 per 100,000
- Australian national rate: approximately 1.2 per 100,000 (RLSSA National Report)
- Non-fatal drownings: 60% above 10-year average
- CALD community representation: ~40% of deaths — highest proportion ever recorded
- Males: 46 of 54 deaths (85%) — consistent with global pattern of ~80%
- All 54 deaths at unpatrolled locations — 0 deaths at patrolled locations
- WHO global rate: ~320,000 drowning deaths per year

Sources: LSV — lsv.com.au/research/victorian-drowning-reports/ | RLSSA — royallifesaving.com.au | WHO — who.int/news-room/fact-sheets/detail/drowning

Activity 1.1 — Calculations & Interpretation

Individual · ~15 min

Question 1.1a — Calculation [2 marks]

Using the population data above, calculate: what PERCENTAGE of the 54 Victorian drowning deaths were male? Show your working. Then compare to the global figure (80%) and explain what the difference suggests.

Working: 46 / 54 x 100 = ____________%

Question 1.1b — Rate Comparison [3 marks]

Victoria's rate = 0.78 per 100,000. National rate = 1.2 per 100,000. Calculate: at the national rate, how many deaths would Victoria expect? What does this tell you about Victoria's performance? Show all working.

Target deaths = 1.2 x 69 (units of 100,000 in 6.9M) = ________

Question 1.1c — CALD Data Analysis [3 marks]

CALD communities = 40% of Victorian drowning deaths, ~28% of population. Are CALD Victorians over-represented, under-represented, or proportionally represented? Show your reasoning. Give TWO specific reasons (not just "language barriers") for this pattern.

Over-represented / Under-represented / Proportional (circle one). Evidence: _______

Question 1.1d — Data Visualisation [3 marks]

Create a bar graph comparing Victorian drowning deaths by location type. Research the approximate figures from the LSV report (lsv.com.au/research/victorian-drowning-reports/). Label axes, include a descriptive title, and write a 2-sentence analysis beneath.

[Attach or draw your bar graph here / on a separate page]

Analysis: ____

2 Beach Flag System — Critical Evaluation

Activity 2.1 — Flag Analysis [Colour required]

Individual · ~10 min

Beyond just identifying flags — analyse WHY each flag exists and WHO it protects.

Flag	Why this flag exists	Who it protects	Limitation of this flag
Red & Yellow			
Single Red			
Black & White			
Yellow + Black Ball			
Orange Windsock			
Purple			

Question 2.1a — Critical Evaluation [3 marks]

A student argues: "The flag system is pointless — people just ignore it anyway." Evaluate this argument. Agree or disagree? Use evidence from Victorian drowning data.

I agree / disagree (circle) because: ________________________________

Activity 2.2 — Rip Current: Physics & Strategy [Colour required]

~12 min

Rip currents can flow at up to 2.5 m/s. The world record 100m freestyle pace is approx. 2.0 m/s.

Question 2.2a — Physics Reasoning [3 marks]

A rip flows at 2.5 m/s. The world record swimmer averages ~2.0 m/s. Explain precisely why even a highly capable swimmer cannot escape a rip by swimming directly against it.

The rip speed (2.5 m/s) is ________ *than the world record (2.0 m/s). Net movement =* ________ *m/s away from shore.*

Question 2.2b — Strategy Evaluation [3 marks]

LSV advises floating and signalling rather than attempting to swim out of a rip. Evaluate this advice from a physiological standpoint — why does energy conservation matter more?

Research extension — new statistic from sls.com.au: ____________________

Source URL: ____

3 Emergency Response — Advanced Application

Deeper Knowledge

- **CPR physiology:** Chest compressions generate ~25-30% of normal cardiac output — enough to keep the brain alive until the heart restarts.
- **4-minute rule:** Brain cells begin dying after 4-6 minutes without oxygenated blood.
- **Bystander effect:** Research shows bystander CPR rates are higher when individuals feel personally responsible.
- **Compression rate:** 100-120 bpm; depth 5-6cm; allow full chest recoil between compressions.

Source: St John Ambulance Vic — stjohnvic.com.au | Heart Foundation — heartfoundation.org.au

Activity 3.1 — Advanced Emergency Analysis

Individual · ~15 min

Question 3.1a — Physiological Explanation [3 marks]

Explain why starting CPR within 4 minutes is more important than starting after 8 minutes, even if professional help arrives at both times. Use: cardiac output, oxygenated blood, brain cells, irreversible damage.

Question 3.1b — Bystander Effect [3 marks]

The "bystander effect": people in a crowd are less likely to help because they assume someone else will. Explain specifically how you would overcome this at a beach incident to ensure CPR starts quickly.

Instead of calling out to "someone," I would: ___________________________

Extended Scenario:

A group of 6 students are at an unpatrolled beach. One student is unconscious after 3 minutes in the water. You have a phone. The nearest lifeguard is 2km away. The nearest AED is 400m away.

Question 3.1c — Multi-Step Decision Making [5 marks]

Describe the exact sequence of actions for the first 5 minutes. Assign roles to the 6 people present.

Person	Assigned role	Why this role?
Person 1 (you)		
Person 2		

Person	Assigned role	Why this role?
Person 3		
Persons 4-6		

If not breathing, the priority order changes because: ___________________

4 Risk Analysis, Social Factors & Decision-Making

Activity 4.1 — Social Determinants of Drowning Risk

Individual · ~15 min

Extension context:

Drowning is not equally distributed across the population. Social factors — including gender norms, cultural background, and socioeconomic status — significantly influence who drowns and why. These factors often matter more than swimming ability alone.

Question 4.1a — Gender Analysis [3 marks]

85% of Victorian drowning victims in 2023-24 were male. Explain what "masculinity norms" are, how they increase drowning risk specifically, and propose ONE strategy a school could use to address this.

Question 4.1b — Hazard Mapping with Justification [4 marks]

For River/Dam and Unpatrolled Beach: identify 3 hazards each. Rank H/M/L. Explain the MECHANISM by which each hazard causes drowning — not just that it is "dangerous."

Environment	Hazard	Rank	Mechanism (how it causes drowning)
River / Dam			
Unpatrolled Beach			

Question 4.1c — Peer Pressure: Psychological Analysis [3 marks]

Peer pressure is strongest when: (a) the group is watching, (b) there is a clear dare, (c) refusing seems like weakness. Design a TWO-STEP strategy for resisting a dare to swim at an unpatrolled location. Your strategy must address all three psychological triggers.

5 Inland Waterways — Physiology, Research & Rescue

Activity 5.1 — Cold Water Shock: Physiological Analysis

Individual · ~15 min

Question 5.1a — Physiological Sequence [3 marks]

Describe the complete physiological sequence when a person enters water at 12 degrees C unexpectedly. Use: gasping reflex, hyperventilation, peripheral vasoconstriction, cardiac stress, swimming failure. Explain why this is dangerous even in shallow water.

Question 5.1b — 1-10-1 Rule Application [2 marks]

Explain the 1-10-1 cold water survival rule and describe how you would apply it if you fell into Lake Eildon in December.

Minute 1 (breathing control): ______________________________

Question 5.1c — Research Investigation [3 marks]

Visit paddle.org.au and park.vic.gov.au. For ONE specific Victorian river or lake, research: (a) specific hazards, (b) seasonal factors, (c) any recent safety notes.

Waterway: ______________________________

Question 5.2 — Throw Rescue: Design Your Own Protocol [3 marks]

You are kayaking 15m from shore. One person's kayak overturns in cold water. Design a 3-step rescue protocol using RLSSA hierarchy, accounting for the 15m distance, cold water, and remote location.

Question 5.3 — Alcohol: Multi-Factor Risk Analysis [3 marks]

Construct a multi-factor risk argument for why alcohol + cold Victorian river water is a particularly dangerous combination. Address at least FOUR physiological or behavioural mechanisms, citing the relevant body system for each.

6 Personal Action Plan & Communication Campaign

Extension Assessment Task — Communication Campaign

Choose ONE option. All require evidence-based argument, at least THREE credible sources with URLs, and a sophisticated understanding of your target audience.

Option A — Evidence-Based Social Campaign (targeted at a high-risk group):

Design a complete 3-slide social media campaign targeting ONE specific high-risk group (e.g. young males, CALD communities, inland waterway users, alcohol-affected swimmers). Each slide must: address a specific barrier for that group, cite a statistic from LSV/WHO data, and include a measurable call to action. Justify your design choices in 100 words.

Option B — Policy Brief to a Government Minister:

Write a 400-500 word policy brief to the Victorian Minister for Sport recommending TWO specific government actions to reduce CALD drowning disproportionality. Must: cite evidence, acknowledge counter-arguments, and propose a measurable success metric for each recommendation.

Option C — School Safety Audit Report:

Conduct a water safety audit of your school: what does it currently provide? What are the gaps? Produce a structured 400-word report benchmarked against Curriculum 2.0 requirements (VCHPEP126, VCHPEM146) and LSV classroom resources. Include a priority matrix.

Option chosen: ________ | Target audience / addressee: ______________________________

Central argument / key claim: ____

Source 1 (URL): ____

Source 2 (URL): ____

Source 3 (URL): ____

Evidence-based statistic I will use: ____

Extension Self-Assessment — Rate yourself:

Skill / Knowledge Area	Developing	Solid	Can teach it
Calculate and interpret drowning rate statistics (per 100,000)	■	■	■
Explain the physics of rip currents numerically	■	■	■
Explain CPR physiology and the 4-minute rule	■	■	■
Analyse social determinants of drowning risk	■	■	■
Apply the 1-10-1 cold water rule to a scenario	■	■	■
Produce an evidence-based policy argument	■	■	■